

Wireshark for SCADA

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

Wireshark is the primary tool for ICS protocol analysis, incident investigation, and security verification in operational technology networks. This course covers capture techniques using SPAN ports and network taps, display filter construction, and dissector-level analysis of Modbus, DNP3, and OPC UA traffic. Advanced topics include tshark scripting, remote capture over SSH, and Lua dissectors for proprietary protocols.

PREREQUISITES

Completion of at least one protocol course (Modbus, OPC UA, or DNP3) is strongly recommended.

COURSE TOPICS

1. **Introduction to Wireshark:** Three-pane interface, dissector architecture, ICS protocol support, and the difference between tshark and dumpcap.
2. **Capture Techniques:** SPAN ports, network taps, promiscuous mode, ring buffers, pcap/pcapng formats, and remote capture over SSH.
3. **Display Filters:** BPF capture filters, Wireshark display filter syntax, SCADA-specific expressions, colorization rules, and Follow TCP Stream.
4. **Protocol Dissectors:** Dissector mechanics, Decode As for non-standard ports, Wireshark profiles, and Statistics tools for traffic pattern analysis.
5. **Modbus Analysis:** MBAP header fields, transaction ID tracking, write command identification, and exception response detection in captures.
6. **DNP3 Analysis:** Frame structure, unsolicited response identification, IIN bit monitoring, and Secure Authentication v5 handshake verification.
7. **OPC UA Analysis:** Session establishment, SecureChannel negotiation, security mode verification, and subscription traffic profiling from captures.
8. **Security Analysis:** Traffic baseline establishment, anomaly detection, scanning identification, and incident response capture procedures.
9. **Advanced Techniques:** tshark scripting, SSH remote capture, Lua dissectors for proprietary protocols, and packet replay with tcpreplay.
10. **Protocol Lab:** Modbus poll overload diagnosis, Exception Code 02 trace, OPC UA security audit, rogue master detection, tshark alert script.

LEARNING OUTCOMES

- Capture ICS protocol traffic using SPAN ports and network taps without network disruption.
- Write Wireshark display filters targeting Modbus, DNP3, and OPC UA traffic patterns.
- Verify OPC UA security mode configuration from a packet-level session capture.
- Identify event buffer overflow, exception responses, and authentication handshakes in captures.
- Automate protocol monitoring with tshark scripts for recurring or unattended analysis tasks.

REQUIRED SOFTWARE

- **Wireshark and tshark** (free, cross-platform) — packet capture, display filter construction, and scripted analysis for all lab exercises.
- **Sample PCAP files** (provided with course materials) — pre-recorded Modbus, DNP3, and OPC UA captures for dissector exercises.